

MTH 377/577 Convex Optimization

End-semester exam 2025-26

Instructions: Q1,2,3,4,5 are for all students. In addition, MTH 577 students are required to also answer Q6. Time= 1 hr 30 minutes.

1. Consider the following optimization problem:

$$\begin{aligned} \min \quad & x_1^2 + x_2^2 + x_3^2 + x_4^2 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 = 1 \\ & x_4 \leq k \end{aligned}$$

Let $k \geq 0$. Find the optimal value(s) of the above using KKT conditions. Write down all the steps. (4)

2. State whether the following statements are true or false and provide a formal explanation for your answer. [no marks for only writing true or false without explanation]

(a) All affine sets are convex. (2)

(b) All proportional allocations are envy free. (2)

(c) Let A be a compact subset of $\mathbb{R}^n$. A continuous function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ has (i) a maxima or a minima on A ; (ii) a fixed point. (3)

(d) KKT is sufficient condition for optimality when the objective function f_0 is non-convex. (3)

3. Let $A = \bigcap_{k=1}^n H_k$ where H_k , $k = 1, 2, \dots, n$ are half spaces. Suppose A is closed. Is A convex? Show formally. (3)

4. Consider a vector valued function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and a function $V(f(x))$ that assigns a real value to each $f(x)$, $x \in \mathbb{R}^2$. Let $\succeq$ be defined over $\mathbb{R}_2$. Suppose you want to minimize V .

(a) Given the constraint $x_1, x_2 \in [0, 3]$, write down the minimization problem for generic $V(f(x))$. (2)

- (b) Now, let $f(x) = \{(x_1, x_2) | x_1^2 + x_2^2 = k, k \geq 0\}$ and $f_i(x) = x_1 + x_2 \leq 2$; $x_1, x_2 \geq 0$. Suppose you are minimizing $V(f(x)) = k$. What is the set of minimal elements? Are these pareto optimal? (3)

5. Consider the following optimization problem:

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{s.t.} & f_i(x) \leq u_i, i = 1, 2, \dots, m \\ & h_i(x) = v_i, i = 1, 2, \dots, p \end{array}$$

where $x \in R^n$ Suppose strong duality holds and f_0, f_i, h_i are all continuously differentiable. Suppose u_i increases by $\alpha > 0$ and v decreases by $\beta > 0$. How will the optimal value of f_0 change? Show formally. (3)

6. [Only for MTH 577] Consider the following optimization problem:

$$\begin{array}{ll} \min_{x \in R^n} & f_0(x) \\ \text{s.t.} & f_i(x) \leq 0, \quad i = 1, 2, \dots, m, \\ & h_i(x) = 0, \quad i = 1, 2, \dots, p. \end{array}$$

Write down the Lagrangian dual function for the above problem and show that:

(a) It is always concave. (2)

(b) It is a lower bound for the optimal value of f_0 . (3)